

Beyond Save Percentage: A Situational Expected Goals Differential Framework for Goaltender Evaluation in SHL

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Abstract. Goaltenders are widely regarded as the most influential position in hockey, serving as the final line of defense and often shifting game momentum under high-pressure conditions. In this study, we propose a goaltender-focused extension of the expected goals (xG) framework that incorporates situational factors such as shot distance, shot angle, rebound status, rush chances, game score state, and time since last event. Using 541,802 events from Sportlogiq tracking data, we utilize Markov chains to predict event sequences and game states. From these, we develop a Situational Expected Goals Differential (SEGD) metric to assess goaltender performance relative to context-adjusted expectations. Our results reveal the different ways that the SEGDMetric can be used when assessing performance during a game.

Keywords: Expected Goals (xG), Markov Chains, Spatial Analysis

1. Introduction

Elite goaltending is widely recognized as a critical factor for success in the National Hockey League (NHL) playoffs. Teams that advance deep into the postseason almost invariably rely on consistently strong goaltender performances, often “stealing” games by outperforming expectations under intense pressure. For example, during the 2023 Stanley Cup Playoffs, Adin Hill of the Vegas Golden Knights posted a .932 save percentage and a 2.17 goals-against average (GAA), playing a pivotal role in their championship run. Similarly, in 2019, Jordan Binnington of the St. Louis Blues delivered a standout performance with a .914 save percentage and a 2.46 GAA, helping secure the franchise’s first Stanley Cup. These cases underscore the vital role of elite goaltending in determining playoff outcomes and motivate deeper investigation into rigorous, context-sensitive evaluation methods.

Historically, goaltender assessment has relied on aggregate statistics such as save percentage and GAA. However, these measures often fail to account for shot quality and game context, limiting their ability to isolate individual performance from team effects. Several studies have sought to address these shortcomings. A study from Southern Methodist University (SMU) critiques traditional metrics and proposes context-adjusted alternatives like expected goals against and adjusted save percentage, which control for shot type and shot difficulty. Similarly, Michael E. Schuckers' work on goaltender analytics introduces adjusted save percentage and Defense Independent Goaltender Rating (DIGR) to better account for shot quality and reduce team bias in performance evaluations. Further, research by Gavin Abrie and James Enos at the United States Military Academy explores advanced metrics such as Goals Saved Above Expected (GSAX), incorporating shot type, location, and other situational variables. Building on these foundations, our work aims to refine goaltender evaluation further by developing a context-adjusted, event-level model that quantifies performance relative to both shot quality and game state.

2. Background

A foundational understanding of several domains is necessary to fully engage with the methods and findings presented in this project. These include knowledge of ice hockey game structure and terminology, hockey analytics and advanced metrics, data science principles, Markov chain modeling, and statistical modeling and evaluation.

First, familiarity with the rules of ice hockey, player positions — particularly the role of the goaltender — and standard in-game events such as shots, faceoffs, passes, and blocks are essential. This understanding provides the necessary context for interpreting how game situations and player actions interact, forming the basis for the event-level analysis conducted in this study.

Second, a basic understanding of Markov chains is important for appreciating how sequences of game events can be modeled probabilistically. Markov chains provide a useful framework for representing the progression of plays in hockey, where the likelihood of future states (e.g., shots against, goals allowed) depends on the current state of play and is incorporated into our evaluation of goaltender performance.

Finally, a working knowledge of statistical modeling and evaluation is required. This includes familiarity with key concepts in sports analytics, such as the construction and interpretation of context-adjusted performance metrics. Understanding how to compare observed outcomes (e.g., goals allowed) to model-based expectations (e.g., expected goals against) is central to assessing goaltender performance in a statistically rigorous manner, and forms a core component of our methodology.

3. Algorithms

To develop the SEGD metric, we first conducted extensive exploratory data analysis to better understand the population of NHL goaltenders and the distribution of shot events under different game contexts. This initial step was critical to ensure that the metric

would meaningfully capture variations in goaltender performance relative to situational factors rather than aggregate averages.

The core algorithm for SEGD construction involved adjusting traditional expected goals (xG) models to account for key contextual variables. Specifically, we calculated adjustment weights for score differential, period time remaining, and manpower situations (e.g., power plays or penalty kills). Each available xG value was multiplied by these adjustment weights to generate a context-sensitive SEGD value for every shot event.

In addition to these adjustments, we introduced sequence context factoring, where sequences of prior events leading to a shot (such as zone entries, turnovers, or sustained pressure) were used to further modify the expected goal probability. Recognizing the importance of team effects, we also incorporated team defense quality — quantified as a weighted sum of entry suppression, turnover rates, and shot suppression — and opponent offensive strength into the SEGD calculation.

By building SEGD in this manner, we aimed to overcome limitations in traditional goaltender evaluation metrics that often ignore game context and team effects. Compared to prior work, such as djusted save percentage and Goals Saved Above Expected (GSAx), our approach provides a more nuanced understanding of goaltender performance by embedding dynamic game conditions into the evaluation framework. Overall, our findings suggest that factoring in both game situation and event sequences can reveal significant differences in goaltender performance that traditional metrics fail to capture.

4. Results

After computing the SEGD metric, we conducted a series of analyses to evaluate the validity of our model. A key step involved examining the relationship between SEGD and expected goals (xG). A strong linear correlation between these metrics would indicate that the model appropriately assigns greater value to saves made on higher-quality scoring chances. Figure 1 presents the results of this analysis.

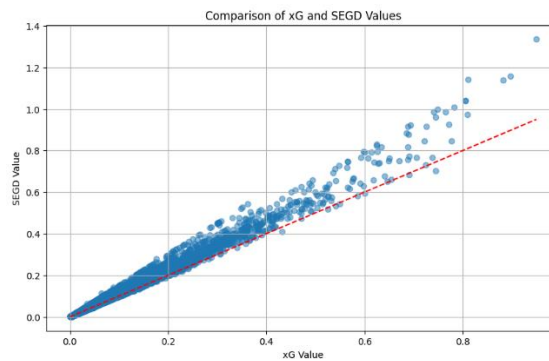


Fig. 1. Expected Goals (xG) and Save Expected Goals Differential (SEGD)

Figure 1 illustrates a strong positive correlation between Expected Goals (xG) and Save Expected Goals Differential (SEGD), supporting the hypothesis that saves should be weighted by shot difficulty. The data show a largely linear trend, with most shots concentrated in the lower xG range (0.0–0.3), indicating that goalkeepers typically face lower-probability attempts. At higher xG values (greater than 0.5), SEGD variance increases, and the slope steepens, reflecting the greater value of saves on high-danger chances. Several notable outliers at the upper end (xG 0.8–0.9, SEGD 2.5–3.3) further validate the model’s sensitivity to critical saves.

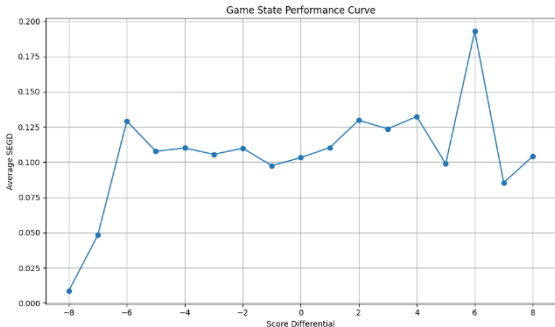


Fig. 2. Relationship between SEG D and Score Differential

Figure 2 reveals a complex, non-monotonic relationship between score differential and Save Expected Goals Differential (SEGD), shaped by established score effects. At extreme negative differentials (–8), low SEG D values (around 0.008) likely reflect high-risk offensive strategies by trailing teams, leading to routine, low-value saves. As differentials moderate (–6 to +4), SEG D stabilizes (0.095–0.130), corresponding to more balanced shot quality. A sharp peak at +6 (SEGD ≈ 0.193) aligns with the well-documented rise in high-danger shots faced leading teams, as opponents adopt increasingly aggressive tactics. Beyond +7, SEG D declines, suggesting reduced offensive pressure as games become less competitive. These findings reinforce how game state alters shot distributions and the value of goaltender performance.

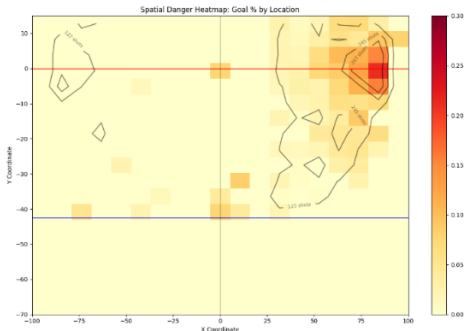


Fig. 3. Spatial Heatmap of Goal Probability by Shot Location and Shot Volume Distribution

The spatial danger heatmap shown in Figure 3 provides quantitative evidence of the relationship between shot location and goal probability in ice hockey. Using a grid, visualization highlights statistically significant spatial patterns in offensive effectiveness. The color gradient reflects goal percentages, from low ($\leq 5\%$) to peak efficiency (30%). A pronounced high-danger zone emerges directly in front of the net, where conversion rates reach 25–30%, aligning with intuitive expectations. Goal probability declines sharply with increasing distance or shooting angle. Overlaid contour annotations reveal major shot clusters, including 257 shots in the right offensive zone aligned with high-danger areas, and a secondary cluster of 122 shots in the lower defensive zone, likely from transitional play. The asymmetry between positive and negative x-coordinates hint at potential defensive structural differences worth future investigation. Overall, Figure 3 offers a reliable framework for evaluating goaltender performance and informs strategic decisions on player positioning, translating spatial data into actionable insights.

5. Future Work

Future enhancements to the SEGD metric could involve developing a normalized version, transforming raw SEGD values into standardized scores to facilitate fair comparisons across goaltenders, regardless of differences in shot faced, time on ice, or team defensive environments. Such normalization would not only allow for more equitable evaluations but also improve the metric’s predictive power for future performance, an important consideration for forecasting and personnel decisions. Beyond normalization, future work could explore moving from rule-based adjustment weights to model-based estimations to learn optimal context adjustments directly from the data. These extensions would further refine the metric’s ability to capture the dynamic, sequential nature of play and better isolate individual goaltender contributions from broader team effects.

Here is the link to our code:

<https://github.com/RyanD17/LINHAC-2025/tree/improved-goalie-metrics>

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